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Scale effects in rockfill behaviour

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The particle sizes in large rockfill structures such as dams prevent laboratory testing. The prediction of field behaviour requires the development of models that integrate size effects. A distinct element method model was developed in which grains are characterised by aggregations of a maximum of 14 elementary spherical particles (the resulting particle shape approaches real geometries and allows a reasonable breakage evolution) and the particle breakage criterion involves the subcritical propagation of fissures in the grain. Time effects are included through the velocity of crack propagation, a function of stress state and defect size, which is introduced as a random set of varying lengths. The model was used to simulate the stress–strain response, the evolution of grain size distribution and creep behaviour under oedometric conditions. The model has been used to simulate size effects in the range 0·28–56·0 cm of initial particle size (uniform distributions were tested). Compressibility and creep were partially validated by comparing calculations with test results covering a reduced range of particles. The paper presents the evolution of short-term compressibility and creep indices in terms of particle size. The model is a useful and novel tool with which to extrapolate laboratory results from scaled grain size distributions to prototype dimensions.

KEYWORDS: compressibility; numerical modelling; particle crushing/crushability

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MOTIVATION

Rockfill is often used in large civil engineering structures such as embankments and dams. Particle sizes range from a few millimetres to over a metre. Common D_{50} sizes in rockfill dams are in the range 10–40 cm. Testing such granular aggregate under oedometric, direct shear or triaxial conditions is unreasonable – the largest testing cells described in the literature (Marsal, 1967, 1973; Marachi *et al*., 1969, 1972; Nobari & Duncan, 1972) are capable of testing rockfill aggregates with maximum particle sizes not exceeding 15–20 cm.

Published experimental information on the mechanical behaviour of rockfill is based on tests on scaled grain size distributions. However, rockfill behaviour is grain size scale dependent in a significant way, as demonstrated by tests performed on materials with 'similar' grain size distributions but having a varying mean grain size (Marsal, 1967, 1973; Marachi *et al*., 1969; Alonso *et al*., 2005).

Scale effects affect particle breakage, a key phenomenon controlling all aspects of the stress–strain–time behaviour of rockfill. It is therefore expected that scale effects are present in the constitutive behaviour of rockfill. Since testing real size is impossible in practice, alternative procedures should be derived to extrapolate the known behaviour at a reduced scale to in situ conditions.

In some applications, specific aspects of behaviour are required (e.g. strength, compressibility and collapse upon wetting, and long-term creep behaviour); examples related to dam engineering are given by Alonso *et al*. (2005) and Frossard *et al*. (2012). Rules have been proposed to scale some rockfill properties such as strength (Frossard *et al*., 2012). This is useful, but difficulties arise when considering other aspects of constitutive behaviour. The distinct element method (DEM) (Cundall & Strack, 1979) offers an interesting alternative to simulate size effects provided it is capable of integrating properly grain failure mechanisms. This is the purpose of this paper.

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THE NATURE OF SCALE EFFECTS

A classical explanation of size effects observed in brittle materials is associated with concepts of fracture mechanics. Failure is understood as a consequence of a full propagation of an existing defect or crack. Under a simple tensile stress state (mode I of failure) of intensity σ, crack propagation occurs when the stress intensity factor

$$K = \beta\sigma(\pi a)^{1/2} \tag{1}$$

becomes equal to the material toughness K_c. In equation (1), a is the initial length of the propagating crack and β is a dimensionless factor that embodies the geometry of the particle, the position of the crack, the direction and point of application of loads and the relative size of the crack in relation to the particle size. It is a function of the ratio a/D where D is taken as the average diameter of the particle. The parameter $\beta(a/D)$ was approximated by the expression given in Tada *et al*. (1985) and Oldecop & Alonso (2007) for a disc of a perfectly elastic–brittle material containing a central crack.

If a given rock mass is assumed to host a distribution of cracks of different lengths, the larger the particle under consideration the higher the probability of having large-size cracks within the particle. In other words, the size of the particle and the size of maximum crack are equivalent. In view of equation (1), particle fracture will depend on particle size.

The next step would be to relate the macroscopic behaviour to the scale-dependent particle breakage. This is an 'automatic' outcome of DEM simulation. Therefore, size effects will be obtained if scale-dependent laws are used to simulate particle breakage.

Scale effects in granular assemblies not undergoing particle breakage were investigated by Kuhn & Bagi

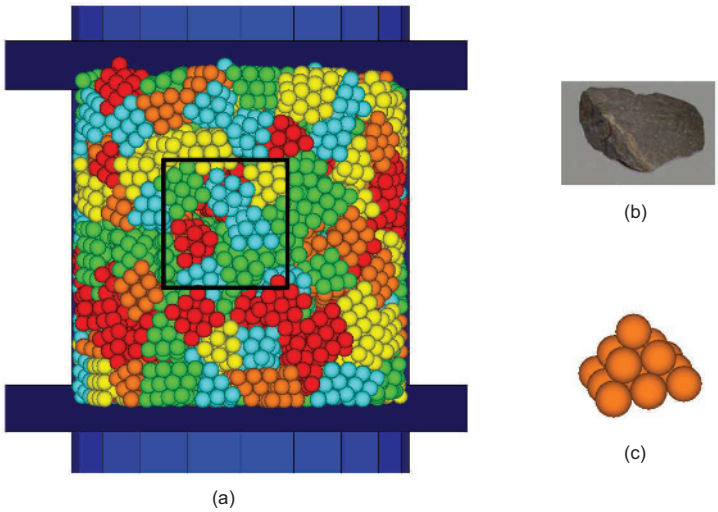


Fig. 1. (a) DEM of oedometer test; sample of 471 macroparticles. (b) Limestone gravel (approximate size 2·5 cm) tested by Ortega (2010). (c) 14-sphere macroparticle

(2009) through DEM simulations of biaxial tests. They found a slight size effect, which was attributed to two sources – the development of localisation bands and 'boundary layer' effects. The first source is attributed to the observation that, as the assembly size increases, deformation is localised within a smaller volume of material. This phenomenon was found to be analogous to the fracture mechanics size effect. The second source was associated with the restriction imposed by the test boundary conditions (caps, lateral restraints) to shear band development. The key point, however, is that calculated size effects were minor. Particle breakage modifies this conclusion substantially.

A DEM MODEL FOR CRUSHABLE PARTICLE AGGREGATES

The numerical model presented here follows previous experimental and theoretical research on the behaviour of rockfill (Oldecop & Alonso, 2001, 2007). The possibilities offered by the DEM (Cundall & Strack, 1979; Potyondy & Cundall, 2004; Itasca, 2008) to investigate scale effects are explored here.

Model particles approximate real shapes by aggregating several spheres as a rigid body. Initial macroparticles (or 'clumps') integrate 14 spheres in the work described here. Figure 1 shows an image of limestone gravel taken from the experimental program referred to later in this paper, the pyramidal shape of the selected macroparticles and the initial arrangement of 471 macroparticles that define the oedometer sample analysed. The selection of the number of spheres simulating a particle was a trade-off between the capability of the assemblage to reproduce a realistic initial shape of particles and the computational cost of the model. A clump of 14 spheres allows a basic three-dimensional pyramidal shape, which was judged adequate. Macroparticles interact through contact friction and normal and shear stiffness coefficients (Table 1).

Table 1. Properties of discrete particles in the DEM

Contact stiffness, k_n: N/m	4×10^6
Shear stiffness, k_s: N/m	4×10^6
Friction coefficient, μ	0·3
Toughness, K_c: Pa.m$^{1/2}$	5×10^6

Defects are attributed randomly to all particles, keeping their maximum length limited by the initial size of particles. Maximum defect length was limited to half the size of the particle. A uniform probability density function was simply selected for the distribution of crack sizes among particles.

Particles will fail if, at some instant during sample loading, $K = K_c$. However, particles may also fail even if $K < K_c$. This is known as a 'subcritical' failure condition, analysed by Wiederhorn *et al.* (1980), Atkinson (1984) and Oldecop & Alonso (2001, 2007) in the context of rockfill behaviour. Fractures propagate in time at a velocity controlled by K and the prevailing relative humidity (RH). The second dependence, which explains collapse deformation upon wetting, is not analysed here. A dry rockfill is assumed in the simulations performed.

Determination of the tensile stress σ necessary to calculate K follows the analysis given by Russell & Muir Wood (2009) and Russell *et al.* (2009). In the vicinity of a spherical particle subjected to a loading stress ρ (force per unit area of application), the tensile stress is given by

$$\sigma = \left(\frac{1}{2} + v - \frac{1+v}{[1+(w/z)^2]^{1/2}} + \frac{1}{2[1+(w/z)^2]^{3/2}} \right) \rho \qquad (2)$$

where v is Poisson's ratio, is z the distance below the centre of the loaded area and w is the radius of a loaded circular area. Macroparticles are, for the purposes of calculating σ, assimilated to spheres of equivalent radius r. The maximum tensile stress is calculated at a distance $z = 0{\cdot}1925r$ under the centre of the loaded area on the boundary of the sphere, for $v = 0{\cdot}25$ and a solid angle 'seen' from the centre of the particle, $2\theta_0 = \pi/18$, which defines the small area of stress application.

The crack propagation velocity V is described by a modified Charles' law (Oldecop & Alonso, 2007)

$$V = \frac{da}{dt} = V_0 (K/K_c)^n \qquad (3)$$

where V_0 is a reference velocity taken as $V_0 = 0{\cdot}1$ m/s in view of the data given by Charles (1958) and $n = 181{\cdot}5$ for dry conditions (RH = 10%). For a given time increment dt, the crack length increment da is calculated following equations (3) and (1). In this work, the time increment has a value of 1000 s and this is related to real time. Crack

Table 2. Dimensions of sample and macroparticles in the numerical simulation of oedometer tests

Sample	Scale	Sample size		Equivalent macroparticle diameter: mm	Initial porosity
		Height: mm	Diameter: mm		
S25	0·1	25	25	2·8	0·5314
S250	1·0	250	250	28	0·5314
S1250	5·0	1250	1250	140	0·5337
S5000	20·0	5000	5000	560	0·5314

lengths are updated at each step in calculations, $a = a_0 + da$, where a_0 is the previous crack length. The particle is assumed to be broken when the updated crack length reaches the mean dimension of the particle.

The macroparticle divides following an arbitrary criterion that takes into account the number of particles integrated in a 'clump' and its pyramidal shape. The following division rule is applied

$$14 \rightarrow 8+6;\ 8 \rightarrow 4+4;\ 6 \rightarrow 3+3;\ 4 \rightarrow 2+2;\ 2 \rightarrow 1+1$$

This tends to divide particles into two new particles of approximately equal shape and size.

The calculation of stresses, determination of failure criteria for particles and the conditions and performance protocol of oedometer tests were introduced in a program developed in FISH language, which is included in the program PFC3D v4 (Itasca, 2008). Any clump that reaches the established failure criterion (length of crack reaching the clump size) divides in the manner described above. The two new particles (clumps) are integrated into the model. Therefore, the number of clumps increases continuously during the 'test'.

Sample dimensions, initial porosity and initial equivalent diameter of particles are given in Table 2. Maximum particle size covers the range 56·0–0·28 cm. Initial porosity is essentially the same in all cases (0·53) to avoid density effects.

MODEL RESPONSE

Short-term compressibility

Model parameters (Table 1) were adjusted to simulate results of real oedometer tests performed on uniform limestone gravel (Ortega, 2010). Ortega tested 40–30 mm, 30–25 mm, 25–20 mm and 20–25 mm gravels having an initial porosity of 0·49 in a cell of 300 mm diameter and 200 mm height. Model calculations for a particle size of 28 mm are compared in Fig. 2 with experimental results for the 30–25 mm range of limestone particles.

The agreement between the calculated and measured deformation–time records demonstrates the capability of the model to reproduce rockfill deformation phenomena. All time records follow a similar pattern – an immediate deformation is followed by a delayed accumulation of strains. Calculated deformations are very similar to the experimental results. The slope of the delayed response is small and it cannot be distinguished in Fig. 2. It is a consequence of the rupture of clumps and the subsequent rearrangement of particles. The calculated slopes given later (see Fig. 6) show the long-term compressibility values. Note that the transition time from short-term compressibility to long-term behaviour is somewhat different when comparing model and experiments.

If deformations for a particular instant (5×10^8 s was selected) are plotted in terms of vertical stress, conventional deformation–stress plots can be calculated (Fig. 3). They show the type of compression behaviour reported by Oldecop & Alonso (2003) for samples of uniformly compacted slate gravel. Scale effects are clearly shown. Larger size particles result in an increase in compressibility index, defined as

$$\lambda = \frac{d\varepsilon_v}{d(\ln \sigma_v)} \tag{4}$$

where ε_v is the vertical deformation and σ_v is the vertical stress.

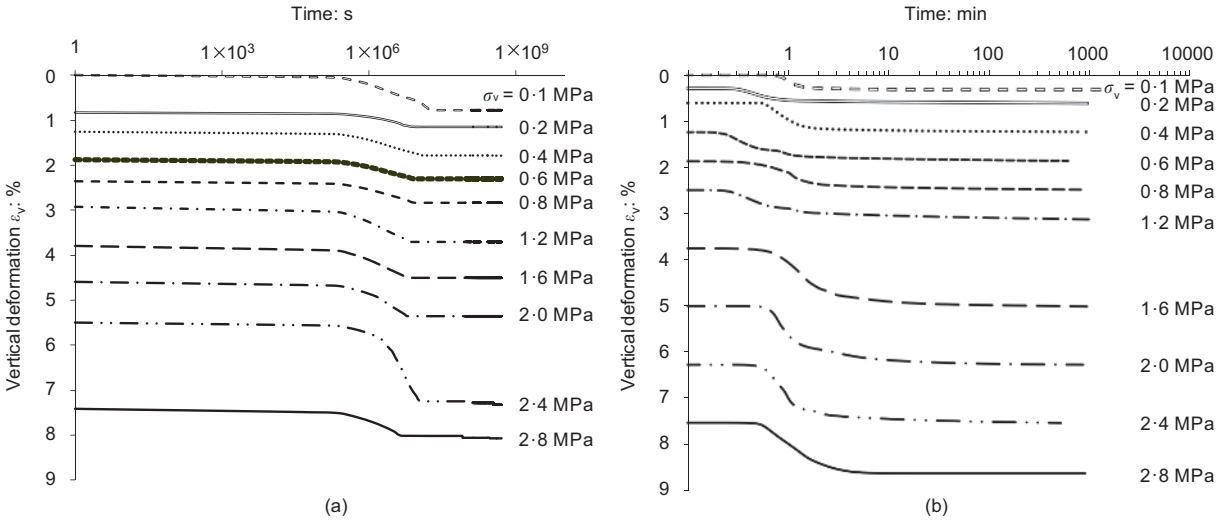


Fig. 2. (a) Calculated deformation–time records under the vertical stresses indicated; equivalent particle size 2·8 cm. (b) Measured deformation–time records in an oedometer test on compacted limestone gravel having equivalent diameter sizes in the range 2·5–3·0 cm (Ortega, 2010)

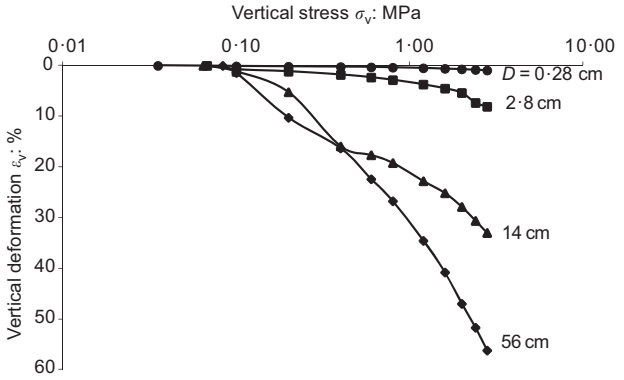


Fig. 3. Calculated compressibility of samples of increasing initial equivalent particle size D

Calculated λ values are shown in Fig. 4. For $\sigma_v = 2$ MPa, the compression coefficient λ ranges from 0·005 to 0·27 when the grain size changes from 0·28 to 56·0 cm (i.e. 200 times larger). The size effects on λ for $\sigma_v = 2{\cdot}8$ MPa are plotted in Fig. 5 along with the λ values measured by Ortega (2010) for three test series. They show the expected scale effect but points are concentrated in a narrow range of sizes. The plotted regression curve is given by

$$\lambda = 0{\cdot}004(D)^{0{\cdot}7} \tag{5}$$

Long-term (creep) compressibility

Long-term deformations are found in the numerical calculations performed (Fig. 2) to be linearly related to log(time). This is also the case of the real experiments reported in the same figure. The long-term compressibility index, defined as

$$\lambda_t = \frac{d\varepsilon_v}{d(\ln t)} \tag{6}$$

increases with applied stress. Equation (6) was applied in discrete form ($\lambda_t = \Delta\varepsilon_v/\ln(t_2/t_1)$) for two times of the strain record: the time corresponding to the total simulation period ($t_2 \approx 5 \times 10^8$ s) and a previous time ($t_1 \approx 5 \times 10^7$ s) also located within the delayed 'flat' response. Figure 6 shows the calculated values in terms of the confining vertical stress for $D_{max} = 2{\cdot}8$ cm. The calculations compare well with experimental results for uniform aggregates of similar size and porosity.

A definite scale effect was also found. This is to be expected because λ and λ_t are known to be closely related

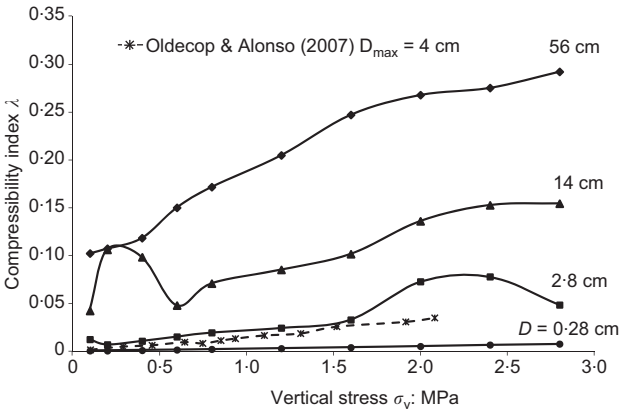


Fig. 4. Calculated compressibility index λ. Also indicated are λ values for an oedometer test on compacted dry quartzitic slate (Oldecop & Alonso, 2007)

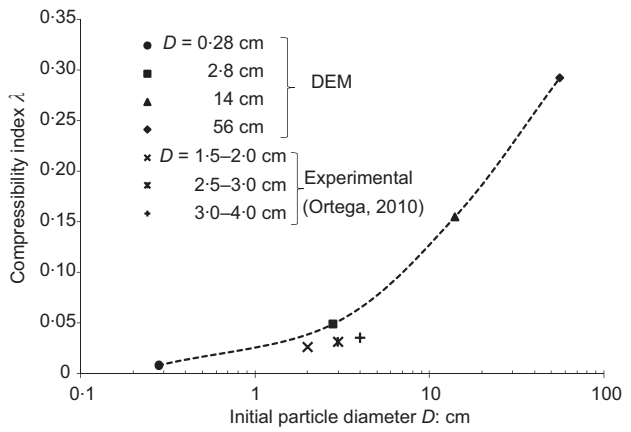


Fig. 5. Variation of compressibility indices with initial particle diameter at $\sigma_v = 2{\cdot}8$ MPa

parameters. Oldecop & Alonso (2007) provided a theoretical explanation for the constant ratio λ_t/λ observed in creep tests on rockfill.

Computed λ and λ_t values exhibit an approximately constant ratio (Fig. 7). Figure 7 also includes experimental data: the range of λ_t/λ values reported by Oldecop & Alonso (2007) for oedometer tests on compacted slate (as well as other data) and the λ_t/λ values reported by Ortega (2010) for limestone gravels. The calculated λ_t and λ values are close to a unique relationship

$$\lambda_t = 0{\cdot}04\lambda \tag{7}$$

which is close to the lower ratio reported by Oldecop & Alonso (2007). This lower ratio corresponds, consistently, to oedometer tests on very dry samples. The values reported by Ortega (2010) span the limits of the λ_t/λ relationship found in other experiments as well as in the present set of calculations.

The model results indicate that aggregates with very large particles follow the λ_t/λ relationship reported for tests on dry specimens of compacted gravels reasonably well.

Particle breakage

Figure 8 shows an example of the evolution of the grain size distribution of the 2·8 cm particle size sample when the

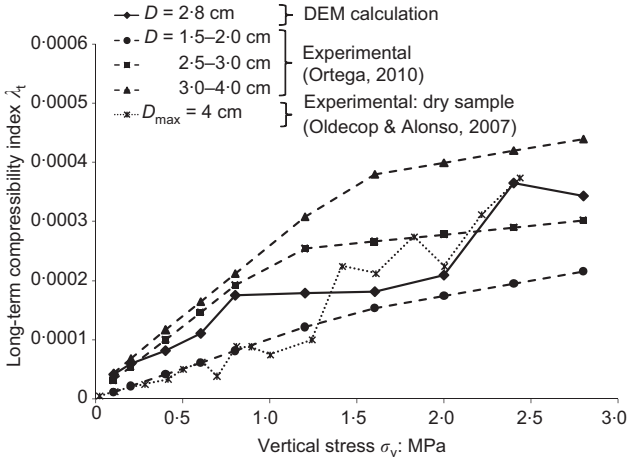


Fig. 6. Calculated long-term compressibility index λ_t (for $D = 2{\cdot}8$ cm) and experimental results based on Ortega (2010) and Oldecop & Alonso (2007)

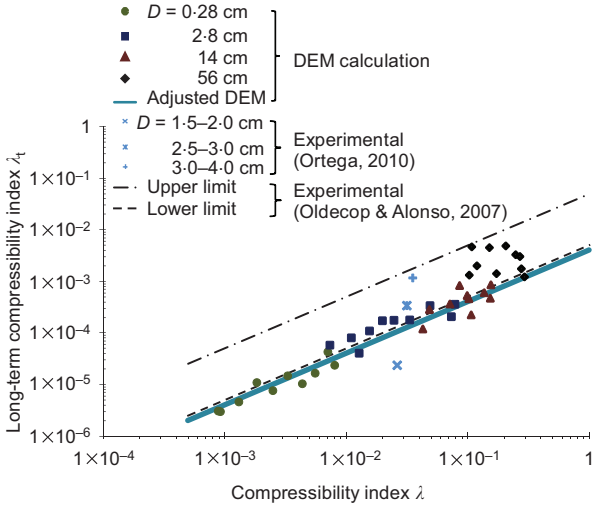


Fig. 7. Correlation between calculated long-term compressibility index λ_t and compressibility index λ and experimental results based on Ortega (2010) and Oldecop & Alonso (2007)

vertical stress reached 2·8 MPa. The final grain size distribution is not a continuous function, which is probably a result of the limited number of particles defining a macroparticle and also the relatively low number of 'clumps' used in the simulation. Nevertheless, the attained distributions allow the calculation of breakage indices (Marsal, 1973; Hardin, 1981). The calculated Hardin index is shown in Fig. 9 in terms of grain size; three values reported by Ortega (2010) are also shown in the figure. Apparently, the calculated breakage indices stabilise beyond a certain grain size (2·8 cm) and even decrease slowly. However, Fig. 9 should be considered a preliminary result subjected to further analysis with increasing numbers of particles.

CONCLUSIONS

Scale effects on rockfill behaviour can be successfully approached through DEM modelling provided a particle breakage criterion incorporating the mechanics of crack propagation in brittle materials is included in the model. This opens the possibility of quantifying scale effects in rockfill, a subject of practical interest.

The model developed simulates particle shapes of roughly pyramidal shape and proposes a particular subdivision law. The subdivision is achieved when a critical defect or crack propagates through the particle. The time to

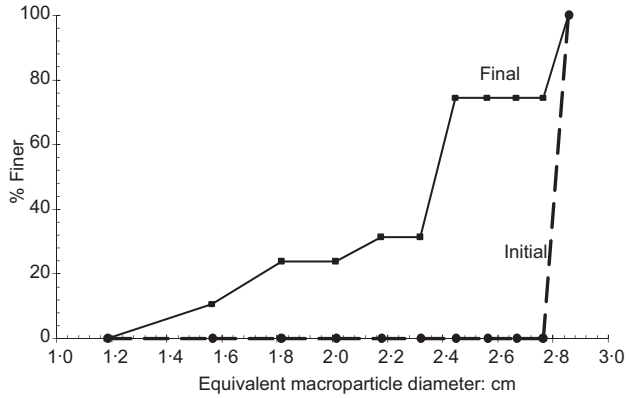


Fig. 8. Calculated grain size distribution of test $D = 2·8$ cm at the end of the test ($\sigma_v = 2·8$ MPa)

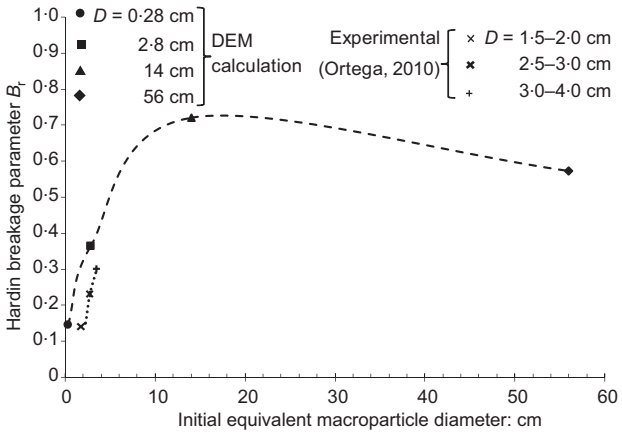


Fig. 9. Hardin breakage parameter (B_r) derived from DEM calculations and B_r values derived from test results (Ortega, 2010)

the final stage of crack propagation is derived from an analysis of the subcritical crack growth under a mode I type of loading, which is accepted as a reasonable mode in granular assemblages in which diametral loading of particles is the dominant mechanism.

The model parameters were calibrated through the results of tests performed on samples with a given grain size distribution. Once this is achieved, the model can be used as a virtual laboratory to explore size effects.

The model is capable of a precise reproduction of long-term (oedometer) tests. Short-term compressibility (λ) increases with particle size, as well as the creep or secondary coefficient. The ratio λ_t/λ was found to be in reasonable agreement with experimental results derived for different granular materials. The paper has also provided information on the evolution of the grain size distribution during loading of specimens for a wide range of particle sizes.

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